**Equations of motion:**

$$x^\top = \begin{pmatrix}\beta & \gamma & \omega\end{pmatrix} \tag{1}$$

$$u^\top = \begin{pmatrix}u_\beta & u_\gamma & \tau_{gen}\end{pmatrix} \tag{2}$$

$$w^\top = \begin{pmatrix}V & \gamma_{wind} & I\end{pmatrix} \tag{3}$$

$$\dot{\beta} = \begin{cases} c_{\dot{\beta}}, & \text{if } u_\beta = 1 \\ -c_{\dot{\beta}}, & \text{if } u_\beta = -1 \\ 0, & \text{if } u_{\dot{\beta}} = 0 \end{cases} \tag{4}$$

$$\dot{\gamma} = \begin{cases} c_{\dot{\gamma}}, & \text{if } u_\gamma = 1 \\ -c_{\dot{\gamma}}, & \text{if } u_\gamma = -1 \\ 0, & \text{if } u_{\dot{\gamma}} = 0 \end{cases} \tag{5}$$

$$\dot{\omega} = \frac{\tau_{ae}(w,x) - \tau_{gen}}{J_t} \tag{6}$$

$$P_{ae}(V,\lambda,\beta) = \frac{1}{2}\rho A V^3 \cos(\gamma_{rel})^3 c_p(\lambda,\beta) I \tag{7}$$

$$F_t(V,\lambda,\beta) = \frac{1}{2}\rho A V^3 \cos(\gamma_{rel})^2 c_t(\lambda,\beta) I \tag{8}$$

$$P_{gen} = \tau_{gen}\,\omega\,\mu \tag{9}$$

$$x_{k+1} = x_k + T_s \dot{x} \tag{10}$$

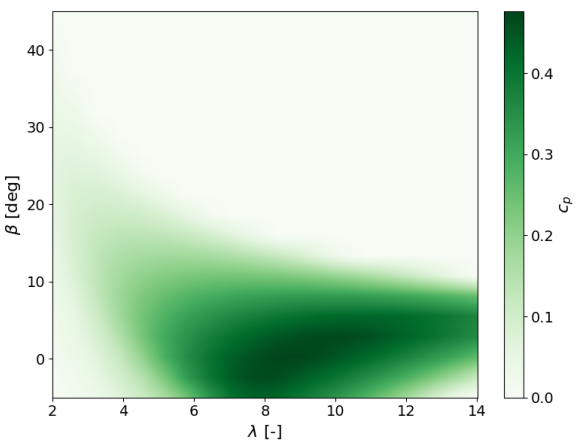



Figure 1: Measurement based interpolated $c_p$

**Local Controller Design:**
The aim of the controller is to track a reference power and yaw angle signal while minimizing the generated drag. In order to do this a model predictive control-like approach is used together with a P controller that essentially inverts the dynamics and uses a linearized approach to approximate the optimal path finding in the trajectory space. The controller is summarized by the flow chart which has pointers to the equation that correspond to each step.

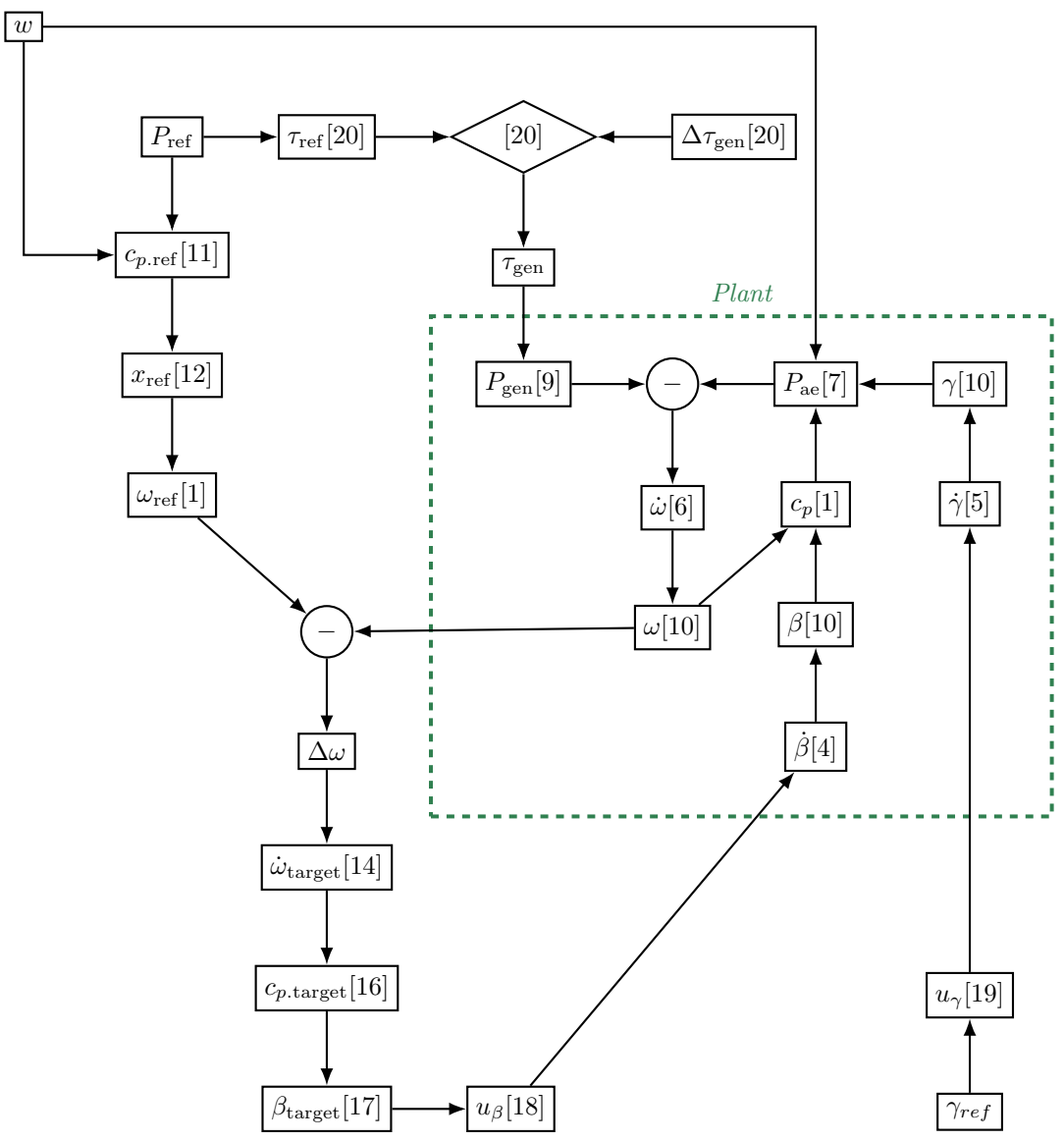
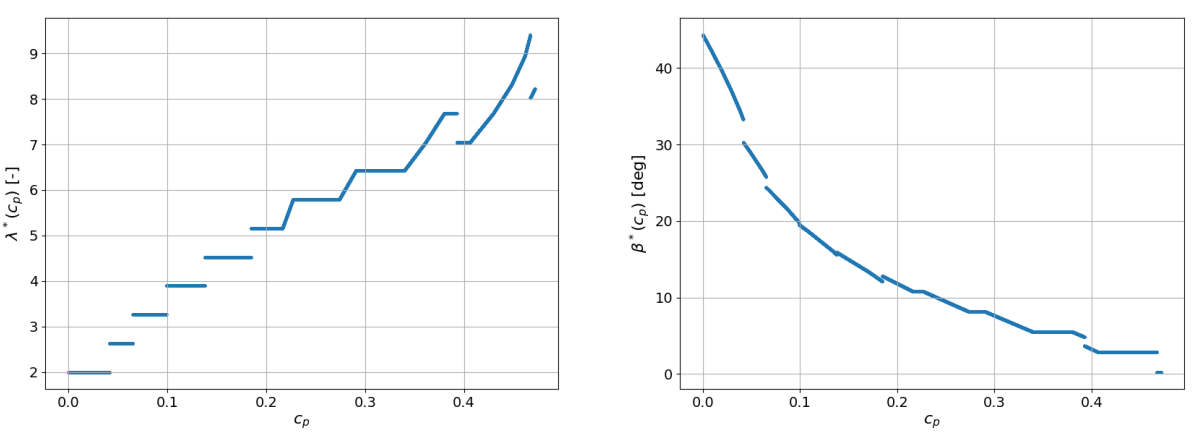



Figure 2: $\beta$ Controller Information Flow

$$c_{p.ref} = max\left(\frac{P_{ref}}{\frac{1}{2}\rho A V^3 cos(\gamma_{rel})^3 I}, c_{p.max}\right) \tag{11}$$

$$\begin{aligned} x_{ref} &= arg\min_x c_t(x) \\ &\quad s.t. \\ c_p(x) &= c_{p.ref} \\ \beta &\in \mathcal{B} \end{aligned} \tag{12}$$




a: $\lambda^*(c_p)$ b: $\beta^*(c_p)$

Figure 3: Mapping $c_p$ to optimal states, and corresponding $c_t$.

$$u^\star(t) = \arg\min_u \frac{1}{T}\int_0^T (P_{gen}(t) - P_{ref})^2 + q \cdot c_t(t, x(t), w(t))^2 dt \tag{13}$$

$$s.t \int_0^T \dot{x}(u(t), t, w(t))dt = x_{ref} - x_0 \dot{\omega}_{target} = p(\omega_{ref} - \omega) \tag{14}$$

$$\tau_{aero.target} = \dot{\omega}_{target} J_t + \frac{P_{ref}}{\omega\mu} \tag{15}$$

$$c_{p.target} = \frac{\tau_{aero.target}\omega}{\rho A V^3 I cos(\gamma_{rel})^3} \tag{16}$$

$$\beta_{target} = \arg\min_{\beta\in\mathcal{B}} (c_p(\beta,\lambda) - c_{p.target})^2 \tag{17}$$

$$u_\beta = \begin{cases} 0 & \text{if } |\beta_{target} - \beta| < \frac{1}{2} c_{\dot{\beta}} T_s \\ \begin{cases} 1 & \text{if } \beta_{target} > \beta \\ -1 & \text{if } \beta_{target} < \beta \end{cases} & otherwise \end{cases} \tag{18}$$

$$u_\gamma = \begin{cases} 0 & \text{if } |\gamma_{\text{target}} - \gamma| < \frac{1}{2} c_{\dot{\gamma}} T_s \\ \begin{cases} 1 & \text{if } \gamma_{\text{target}} > \gamma \\ -1 & \text{if } \gamma_{\text{target}} < \gamma \end{cases} & otherwise \end{cases} \tag{19}$$

$$\tau_{gen} = \begin{cases} \begin{cases} \frac{P_{ref}}{\omega\mu} & if\ \omega \geq 0 \\ 0 & otherwise \end{cases} & if\ c_p(\beta_{target},\lambda) > \frac{P_{ref}}{\rho A V^3 I cos(\gamma_{rel})^3 \mu} \\ \tau_{ae}(w,x) - \dot{\omega} J_t & otherwise \end{cases} \tag{20}$$